

Evaluating Audio Feature Extractors and Multi-Stage Pre-training for Audio-Moment Retrieval

NSF DREAM Fellowship Technical Report

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Abstract—The focus of this research project was to build upon the 2026 DCASE Task 6 challenge to advance language-based audio moment retrieval in long-form audio. Based on multi-modal contrastive representation learning, such as Microsoft Contrastive Language-Audio Pre-training (MS-CLAP) Masked Modeling Duo CLAP (M2D-CLAP), and Frame-Wise Language-Audio Modeling (OpenFLAM), this study replicated the current state-of-the-art benchmark model, identified some of the strengths of the best performing challenge submissions, and evaluated how alterations in audio embedding features impact frame-level temporal retrieval.

Index Terms—Audio Moment Retrieval, Audio-Text Cross-Attention, Temporal Localization

I. INTRODUCTION

Audio Moment Retrieval (AMR) is the task of predicting start and end timestamps, $m = [t_{start}, t_{end}]$ for specific acoustic events within long-form audio, A , based on open-vocabulary natural language queries, Q . Designed by Munakata et al. [1], [2], AMR is a novel task that takes inspiration from gaps in previous work in the language-based audio retrieval domain and current practices for retrieval from long-form contexts in computer vision. Accurate retrieval systems have real-world applications in a wide range of contexts, from multimedia search to surveillance systems.

The current 2026 DCASE baseline applies an Audio Moment DETR (AM-DETR) model, which is developed from standard Query Dependent DETR (QD-DETR) architecture from Video Moment Retrieval (VMR)[1], [2], [3], and utilizes a sliding-window feature extractor based on the pre-trained MS-CLAP [4] with the purpose of creating sequential, cross-modally aligned text and audio embeddings [1], [2]. In DETR-based architectures, such as QD-DETR and AM-DETR, the overall model pipeline consists of two primary components. First, the feature extraction backbone, which maps raw audio and text inputs into sequential embeddings, and second, a transformer detection head, which fuses audio and text representations via query-dependent cross-attention layers using a set of learnable moment query slots via a transformer decoder to predict normalized moment boundaries, confidence scores,

and per-frame saliency ratings [1], [2], [3].

The baseline model is trained in two ways, first on the CASTELLA dataset only and second with a pre-training pipeline using the Clotho-Moment dataset followed by the CASTELLA dataset [5]. This resulted in baseline performance benchmarks of Recall1@0.5 of 23.16% and 25.61% and Recall1@0.7 of 10.32% and 13.59% for CASTELLA-only and CASTELLA + Clotho-Moment trained models, respectively [2], suggesting there is significant opportunity for improvement. Some of the primary challenges of long-form audio include adversarial interference, such as complex background noise, multi-label sound events, temporal extremes, and variations in the temporal scale that require more fine-grained boundary estimation than clip-level classification provides.

To address these limitations, the top three 2026 DCASE Task 6 challenge submissions on AMR achieved Recall1@0.7 of 48.59% on the evaluation dataset. This means that for nearly half of the test queries, the top-ranked (most confident) predicted moment achieved an Intersection over Union (IoU) of 0.7 or higher with the corresponding ground-truth event.

Kibata et al. [6] designed a modified Context-Gated DETR (CG-DETR) framework with M2D-CLAP audio and text feature extractions [7]. The authors note that M2D-CLAP’s self-supervised learning approach that models raw input audio independently of its text pairing has greater generalization compared to MS-CLAP, which may have contributed to the improvements observed in retrieval accuracy [6], [7]. Similarly, Sugawara et al. [8] paired M2D-CLAP and MS-CLAP audio embeddings with MS-CLAP text embeddings processed using a Unified Video COMprehension (UVCOM) framework [9]. UVCOM is a DETR-based network that was developed to solve VMR and Highlight Detection (HD) challenges in computer vision and has been noted by Munakata et al. [5] to have shown performance gains in AMR.

Lastly, Kim et al. [10] used Large-scale Artificial Intelligence Open Network CLAP (LAION-CLAP) [11] as the primary audio-text feature representations with additional selectively integrated acoustic and linguistic features from other popular audio and language encoders and applied a

Quality-Aware Mamba DETR (QAM-DETR) with a frozen audio-language Qwen2.5-Omni-7B LLM verifier as a post-processing re-ranker. The authors note that LAION-CLAP is superior to MS-CLAP due to its ability to utilize feature fusion mechanisms to provide audio and text embeddings in a shared semantic space allowing for better handling of variable-length audio inputs [10], [11].

Kang et al. [12] was the only group to submit a model that replaced the original feature extractor with OpenFLAM while preserving the QD-DETR transformer decoder and prediction head. Interestingly, despite maintaining the original QD-DETR architecture and only replacing the feature encoder, their best performing model was still able to achieve Recall@0.7 of 35.59%. OpenFLAM preserves fine-grained temporal resolution by performing sound event detection at the individual frame level [13]. By targeting temporal misalignment through improving boundary precision using a frozen OpenFLAM encoder and enhancing text variation through an ensemble of paraphrase pooling, saliency-curve candidates, and augmentation-trained students, the authors are able to achieve a relatively lightweight architecture that still performs reasonably well [12].

Across multiple DCASE submissions, DETECTION TRANSFORMER (DETR) based network architectures are used as the detection head. The baseline model demonstrates that AM-DETR/QD-DETR with global MS-CLAP feature extraction suffers from temporal resolution loss due to fixed sliding-window frames and poor boundary regression on short events. This limitation and the strengths of the 2026 DCASE Task 6 submissions highlights the need to systematically isolate the influence of underlying audio embedding encoders via controlled ablation studies to assess how feature granularity impacts frame-level temporal retrieval.

Therefore, to systematically investigate feature representation capacity and pre-training dynamics for AMR, this work addresses the following key research questions:

- **RQ1 (Feature Extractor Impact):** How do specialized self-supervised and frame-level contrastive audio-text representations (M2D-CLAP and OpenFLAM) compare against baseline clip-level contrastive encoders (MS-CLAP) in localizing sound events in long-form audio?
- **RQ2 (Pre-training Transferability):** Does multi-stage pre-training on large synthetic soundscapes (Clotho-Moment) improve query-guided moment boundary detection when fine-tuned on real-world human-annotated datasets (CASTELLA)?
- **RQ3 (Temporal Precision vs. Detection Accuracy):** How do feature extraction backbones impact sound event retrieval under strict ($\text{IoU} = 0.7$) versus compared to coarse ($\text{IoU} = 0.5$) temporal precision requirements for diverse sound events?

II. METHOD

A. Task Definition

The target task is AMR within long-form audio [2], [14]. Given an unconstrained natural language query caption, Q and an audio clip, A , the system predicts the start and end timestamp boundaries $[t_{\text{start}}, t_{\text{end}}]$ of all corresponding acoustic moments in A , along with associated confidence scores. For example, an input text query such as "A car horn blares" for a provided input long audio clip would return the top output moment as predicted relevant time window of $[54.0, 60.0]$, meaning the scene occurs between 54 and 60 seconds.

B. Model Architecture

This system adopts the QD-DETR framework provided as the DCASE Task 6 baseline [2]. Instead of performing a resource- and time-intensive scan over an entire audio clip using a fixed sliding window with independent heuristics, QD-DETR utilizes a cross-attention encoder-decoder transformer architecture. This design directly integrates the natural language text query embeddings into the audio representations through query-guided attention mechanisms.

Through this query-guided prediction task, the network performs multi-modal reasoning to generate context-aware representations connecting audio and text streams based on shared semantic meaning, forcing the model to focus primarily on semantically relevant temporal frames while ignoring unrelated background sound events.

The model optimizes candidate audio moments using a loss function comprising bipartite matching of L1 boundary regression ($\mathcal{L}_{\text{L1}}$) and Generalized IoU (GIoU) loss ($\mathcal{L}_{\text{GIoU}}$), combined with a cross-entropy confidence classification loss ($\mathcal{L}_{\text{CE}}$):

$$\mathcal{L} = \lambda_{\text{L1}} \mathcal{L}_{\text{L1}} + \lambda_{\text{giou}} \mathcal{L}_{\text{GIoU}} + \lambda_{\text{cls}} \mathcal{L}_{\text{CE}} + \lambda_{\text{saliency}} \mathcal{L}_{\text{saliency}} + \lambda_{\text{eos}} \mathcal{L}_{\text{eos}} \quad (1)$$

where predictions are further refined using QD-DETR's highlight detection ($\mathcal{L}_{\text{saliency}}$) and negative pair ($\mathcal{L}_{\text{eos}}$) auxiliary losses [1], [2].

C. Multimodal Feature Representations

Audio and text features from the CASTELLA and Clotho-Moment datasets were additionally extracted using M2D-CLAP and OpenFLAM independently. These extracted representations replaced the default MS-CLAP embeddings used in the baseline system while maintaining the core downstream QD-DETR network architecture unchanged.

1) *Baseline: MS-CLAP:* The baseline model utilizes the MS-CLAP 2023 architecture [4] for feature extraction. The adapted open-source implementation is available online.¹ MS-CLAP employs two distinct encoders, one for text and one for audio, which are trained via contrastive learning on large-scale paired audio-text data. This maps both modalities into a joint embedding space where cosine similarity indicates semantic alignment.

¹<https://github.com/microsoft/CLAP>

Using this pre-trained model, each text query was encoded into a 768-dimensional text embedding, and each corresponding audio file was encoded using 1-second windows with a 1-second hop into a 768-dimensional audio feature sequence (T, D_{audio}) , where $D_{\text{audio}} = 768$. Embeddings were extracted for both the CASTELLA baseline and Clotho-Moment pre-training datasets.

2) *M2D-CLAP*: Unlike standard MS-CLAP, which relies strictly on supervised audio-text pairs, M2D-CLAP incorporates self-supervised learning via randomly injected masking noise in the spectrogram paired with CLAP contrastive loss to create an encoder suitable for zero-shot classification, retrieval, and downstream fine-tuning [7]. For this experiment, the M2D-CLAP 2025 checkpoint was used via the PortableM2D module.²

Audio recordings were resampled to 16kHz mono, zero-padded, and framed into 10-second sliding windows with a 1-second hop size. Inner time dimensions from PortableM2D were mean-pooled across each window to guarantee a 2D temporal feature matrix of shape (T, D_{audio}) where T is the number of time steps (i.e. number of temporal frames the audio clip is divided into) and D_{audio} is the size of the audio feature dimension (i.e. the length of the embedding vector extracted for each frame). Text features were extracted by tokenizing query captions up to a sequence length of 77 tokens and feeding them through a RoBERTa-base transformer model, retaining sequence token representations from the last hidden state without padding.

3) *OpenFLAM*: OpenFLAM was also evaluated, which extends contrastive pre-training from clip-level to fine-grained frame-level alignment, enabling improved open-vocabulary sound event localization [13]. The open-source implementation was adapted using the v1-base model.³

Audio signals were resampled to 48kHz mono and processed using centered 10-second windows with a 1-second hop size. Local features were extracted and mean-pooled across internal patches within each window frame to maintain a consistent (T, D_{audio}) temporal representation matrix. For text encoding, tokens were processed directly through the underlying RoBERTa-base text branch of the OpenFLAM architecture to produce variable-length token embeddings from the unpadded sequences.

III. EXPERIMENTAL SETUP

A. Datasets

The models were evaluated on two datasets specialized for AMR: Clotho-Moment and CASTELLA.

- **Clotho-Moment**: Clotho-Moment is a synthetically generated dataset designed to expose models to large-scale audio recordings with query-moment annotations to enhance temporal alignment. It is constructed by overlapping Clotho audio on Walking Tours background audio

to create simulated audio and text query-audio moment pairs [1].

- **CASTELLA**: The CASTELLA dataset is a manually annotated dataset consisting of long-form audio clips from YouTube recordings with open-vocabulary natural language captions. As a result, the CASTELLA dataset currently serves as the primary benchmark for real-world sound event localization in AMR tasks [5].

B. Model Training Procedure

Each model was either trained only on the CASTELLA dataset or pre-trained on the Clotho-Moment dataset followed by additional fine-tuning on the CASTELLA dataset. Model performance post-training was then evaluated using test queries from the CASTELLA dataset. All experiments were implemented using PyTorch and executed on the High-Performance Computing (HPC) Cluster equipped with NVIDIA A100 GPU using the QD-DETR architecture [2].

1) *Pre-training Stage (Clotho-Moment)*: For initial pre-training, the model was trained on the Clotho-Moment dataset for 200 epochs using a batch size of 32 and an evaluation batch size of 100. An AdamW optimizer with an initial learning rate of 1×10^{-4} and weight decay of 1×10^{-4} were used. Gradient norms were clipped to a maximum of 0.1.

The model architecture consisted of a 2-layer transformer encoder and 2-layer transformer decoder with 8 attention heads, a hidden dimension of $D_{\text{hidden}} = 256$, a feedforward network dimension of $D_{\text{ffn}} = 1024$, sinusoidal position embeddings, and 10 learned object queries. Feature projection layers utilized 2 projection inputs with an input dropout of 0.5 and internal dropout of 0.1. Context modeling operated in `audio_tef` mode (combining audio features with temporal positional embeddings) over maximum sequence lengths of 300 audio frames and 32 text query tokens.

The multi-task objective function was configured with Hungarian matching costs set to 10 for span L1 loss, 1 for Generalized IoU (GIoU) loss, and 4 for classification loss, with loss weight saliency of 1.0 and saliency margin set to 0.2, end-of-sequence coefficient of 0.1, and auxiliary decoding losses enabled [2]. The final loss function coefficients were parameterized as:

$$\mathcal{L} = 10 \cdot \mathcal{L}_{\text{L1}} + 1 \cdot \mathcal{L}_{\text{GIoU}} + 4 \cdot \mathcal{L}_{\text{CE}} + 1.0 \cdot \mathcal{L}_{\text{saliency}} + 0.1 \cdot \mathcal{L}_{\text{eos}}, \quad (2)$$

Best performing model checkpoints and validation logs were saved at 1-epoch intervals.

2) *Fine-tuning Stage (CASTELLA)*: The model was then either only trained on CASTELLA or pre-trained weights from the best checkpoint generated during the Clotho-Moment pre-training stage were fine-tuned using the CASTELLA dataset. The fine-tuning procedure maintained the same optimization parameters (weight decay = 1×10^{-4} , batch size = 32, AdamW optimizer, gradient clipping at 0.1) and training across 200 epochs as were used in the pre-training stage. This helped the model adapt to the CASTELLA dataset's real world YouTube videos, which have complex, unpredictable acoustics and audio clips of all different lengths.

²<https://github.com/nttclab/m2d>

³<https://github.com/adobe-research/openflam>

C. Evaluation Metrics

Following the DCASE 2026 Task 6 evaluation protocol, model performance is evaluated using moment retrieval recall and ranking precision metrics calculated across predicted moments compared against ground-truth temporal boundaries as represented by $[t_{\text{start}}, t_{\text{end}}]$.

- **Recall@ k ($R@k$):** Measures the proportion of ground-truth sound event moments correctly retrieved within the top- k ranked predictions ($k \in \{1\}$). A prediction is counted as a true positive if its temporal overlap with the ground-truth interval satisfies a specified IoU threshold θ :

$$\text{IoU}(\hat{m}, m) = \frac{|\hat{m} \cap m|}{|\hat{m} \cup m|} \geq \theta \quad (3)$$

Performance is reported across loose ($\theta = 0.5$) and strict ($\theta = 0.7$) thresholds, thus $R1@0.7$, to measure broad detection versus fine boundary localization accuracy.

- **Mean Average Precision (mAP):** Evaluates query-guided detection and ranking quality across predicted moments. Average Precision (AP) is computed for each natural language query by ranking predicted moments by confidence score and integrating the precision-recall curve at IoU threshold θ . Results are averaged across all test queries per model to yield the overall mAP score.

IV. RESULTS

Audio moment retrieval performance of the three model architectures, MS-CLAP, M2D-CLAP, and OpenFLAM, are evaluated under two training pipelines, baseline training on CASTELLA dataset only and pre-training incorporating the Clotho-Moment dataset. Performance is measured based on recall at IoU thresholds of 0.5 and 0.7, as well as mAP. Results on the test dataset are summarized in table I.

A. Impact of Clotho-Moment Pre-training

All of the models exhibit a degradation in recall when increasing the strictness of the temporal alignment criteria from an IoU threshold of 0.5 to 0.7 ($R1@0.5$ and $R1@0.7$, respectively). While the absolute rates of decay remain consistent across both training pipelines, the relative retention of top predictions improves when Clotho-Moment pre-training is introduced, with models showing a 49.5 – 54.4% relative drop between $R1@0.5$ and $R1@0.7$ measures under the baseline pipeline, and a 41.4 – 47.7% relative drop under the pre-training pipeline. This suggests that across models, pre-training on temporal audio-moment data, like Clotho-Moment, both improves the model’s performance on event retrieval and results in the model retaining better accuracy in localization despite stricter temporal precision requirements (see figure 1), thus translating in application to better fine-grained boundary localization.

Specifically, incorporating Clotho-Moment pre-training yielded consistent improvements across all feature extraction methods and evaluation metrics. Under the CASTELLA-only baseline, model performance is limited, with the top scores achieved being 27.60% for $R1@0.5$ and 12.02% for mAP

Table I
PERFORMANCE EVALUATION OF MODELS ACROSS TRAINING PIPELINES ON AUDIO MOMENT RETRIEVAL METRICS.

Training Pipeline	Model	R1@0.5	R1@0.7	mAP
CASTELLA Only	MS-CLAP	23.61	11.58	10.25
	M2D-CLAP	27.60	12.59	12.02
	OpenFLAM	24.51	12.37	9.87
CASTELLA + Clotho-Moment	MS-CLAP	29.62	17.37	13.59
	M2D-CLAP	30.69	16.06	13.78
	OpenFLAM	28.58	15.84	12.37

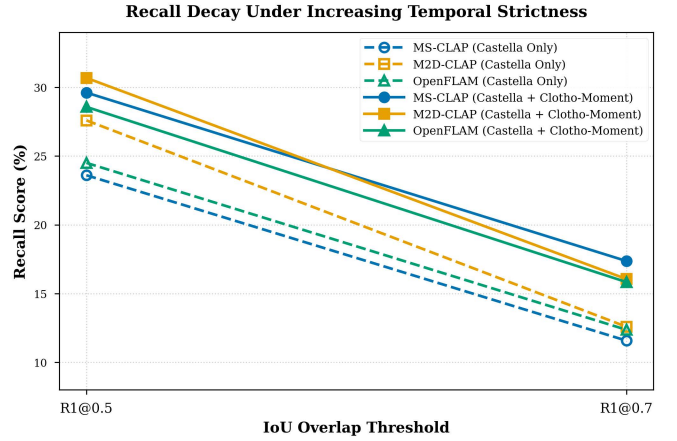


Figure 1. Degradation in recall is noted across models regardless of feature extractors when IoU threshold becomes stricter, however impact on performance is mediated by Clotho-Moment pre-training.

by the architecture using M2D-CLAP for feature extraction. The introduction of Clotho-Moment pre-training provided a substantial boost, with $R1@0.5$ increasing by an average of +4.39 percentage points across models, $R1@0.7$ improving by an average of +4.24 percentage points, and mAP improving by an average of +2.53 percentage points. This supports that domain-specific moment pre-training provides critical temporal alignment information that cannot be fully captured by generic audio-text datasets alone.

B. Impact of Audio-Text Feature Encoders

Among the evaluated architectures, MS-CLAP and M2D-CLAP generally outperformed OpenFLAM. M2D-CLAP exhibits the best overall performance, leading the CASTELLA-only baseline across all metrics and achieves the highest overall scores for $R1@0.5$ and mAP when pre-trained on Clotho-Moment. MS-CLAP demonstrates the highest relative gain from extended pre-training. While it yields the lowest baseline recall scores, Clotho-Moment pre-training results in a +5.79 to +6.01 percentage point improvement. Of note, at the stricter temporal overlap threshold of $R1@0.7$, MS-CLAP achieves the overall highest retrieval score, outperforming M2D-CLAP and OpenFLAM. Lastly, OpenFLAM demonstrated competitive baseline scores on the CASTELLA-only dataset, but displayed less gains from Clotho-Moment pre-training compared to MS-CLAP

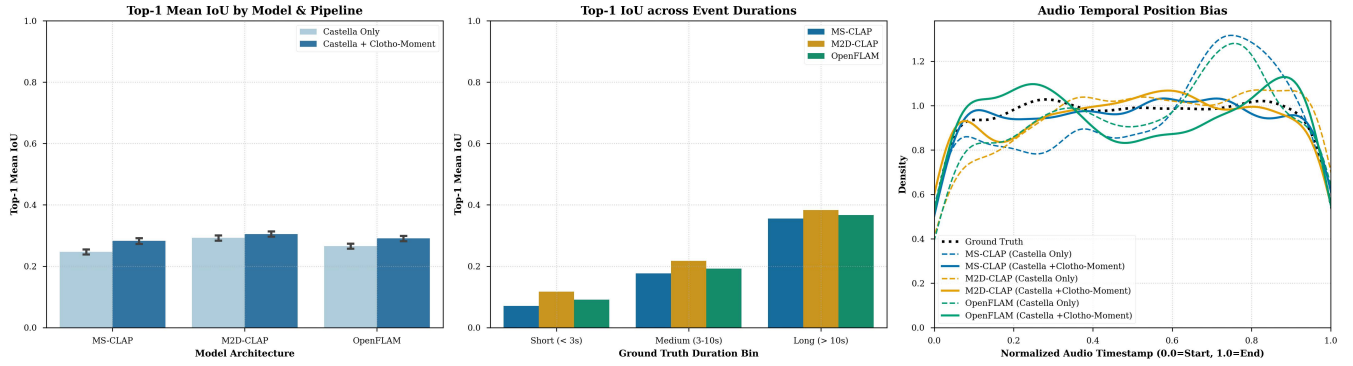


Figure 2. Comparative analysis of sound event localization across model architectures and training pipelines. (Left) Overall top-1 mean IoU across model architectures, comparing single stage (CASTELLA Only) vs multi-stage (CASTELLA + Clotho-Moment) pre-training. Error bars indicate standard error. (Middle) Model performance across ground truth audio moment duration bins (Short: < 3 seconds, Medium: 3 – 10 seconds, Long: > 10 seconds) collapsed across pre-training pipelines. (Right) KDE of predicted normalized event center timestamps (0.0 = start, 1.0 = end) compared against ground truth distribution.

and M2D-CLAP.

C. Characterization of Sound Event Localization

Figure 2 characterizes temporal resolution based on the top-ranked predictions from each of models. Although M2D-CLAP achieved the highest overall performance, MS-CLAP and OpenFLAM showed the largest absolute improvement in localization accuracy when provided with Clotho-Moment pre-training data. Furthermore, all models performed substantially better when target audio moments were longer in duration compared to shorter events. Top-1 Mean IoU scores consistently worsen as duration shortens, indicating that all models struggled to maintain temporal precision as the audio cue becomes more transient.

Furthermore, when looking at the temporal position of predictions to characterize the probability density of sound event midpoints along a normalized audio timeline in comparison to the ground truth, position bias patterns begin to emerge. MS-CLAP and OpenFLAM models trained on CASTELLA-only displayed a late-occurring position bias. They under-predict audio moments that occur in the first approximately 60% of the audio clip and then heavily over-predict audio moments that occur near the end, with a peak around the 80% mark. This suggests that MS-CLAP and OpenFLAM trained only on CASTELLA likely learned a shortcut that sounds happen near the end of the audio file.

As already suggested from earlier findings, the addition of Clotho-Moment in the pre-training pipeline helps to regularize against bias. For example, MS-CLAP and M2D-CLAP curves shift away from the spike at the 0.8 normalized timestamp and more closely follow the ground truth curve, particularly between 0.3 and 1.0. This further demonstrates that pre-training helps the model learn to evaluate the full temporal context rather than just anchoring to specific audio frames.

Of note, M2D-CLAP displays strong positional stability even in the CASTELLA-only baseline, which explains why it generally outperformed compared to other training architectures. Furthermore, this reduced volatility across training pipelines may support that M2D-CLAP is a reliable baseline

model for feature extraction in the underlying network architecture.

Lastly, OpenFLAM with Clotho-Moment pre-training exhibited a bimodal edge bias due to favoring the early and late boundaries of the audio clip, displaying a high density at 0.0 – 0.3 and 0.8 – 1.0 and a low density between 0.3 – 0.7. The addition of pre-training successfully attenuated the late-occurring position bias originally noted for OpenFLAM trained on CASTELLA-only. However, now the model has been pushed towards over-predicting sound events near both boundaries of the audio clip while missing true mid-clip audio moments.

V. DISCUSSION

This ablation study demonstrates that the choice of feature extractor influences audio moment retrieval performance through impacts on sound event localization and temporal prediction behavior. Across all architectures, M2D-CLAP paired with multi-stage pre-training on Clotho-Moment followed by CASTELLA datasets achieved the best overall performance. M2D-CLAP’s masked autoencoder model constructs robust frame-level acoustic representations which likely contribute to the superior localization performance observed. On the other hand, MS-CLAP may be better suited for classification of transient audio cues and global representations, but its limitations in temporal resolution and sensitivity to local spectral noise has potential to degrade its temporal localization and boundary prediction accuracy. OpenFLAM uses both clip-level and frame-level representations to achieve fine-grained alignment for open-vocabulary classification and temporal localization, respectively. However despite strong framewise contrastive representations, OpenFLAM exhibited a tendency toward boundary prediction clustering.

The addition of Clotho-Moment pre-training functioned as a positional regularizer across all the models. In single-stage CASTELLA-only baselines, models like MS-CLAP and OpenFLAM displayed late-occurring position bias, clustering around the 0.8 timestamp of normalized audio clips. Multi-stage pre-training reduced this bias, shifting predicted event

midpoints back towards the ground truth distribution and reducing false positives near audio clip boundaries. Therefore in future work, it may be of interest to further explore training duration and fitting generalization as the models explored here achieved relatively high performance on the Clotho-Moment dataset during pre-training.

Additionally, due to embedding dimension constraints and memory limits, there were notable differences in technical design between MS-CLAP, which required a 1-second sliding window approach, and M2D-CLAP and OpenFLAM, which used a broader 10-second sliding window. The 1-second temporal resolution of MS-CLAP offers structural advantage in pinpointing precise temporal boundaries and combats MS-CLAP’s limitations in localization, but causes it to lack broader global context and leaves it susceptible to interference from localized audio artifact. In contrast, the 10-second sliding window of M2D-CLAP and OpenFLAM capture rich global acoustic context, enabling better overall sequence understanding, but this broader context tends to weaken emphasis on short, transient sound events (< 3 seconds) within downsampled time frames.

Despite the improvement in performance noted from multi-stage pre-training, a substantial performance drop still persists between $R1@0.5$ and $R1@0.7$ across every evaluated setup. While models regularly achieved broad localization, being able to correctly identify the general region of a sound event, they consistently struggled with fine boundary regression, thus failing to identify the exact onset and offset timestamps within narrow temporal tolerances. It is possible that this limitation cannot be resolved with feature extraction modifications alone and that it can be attributed to a weakness of the underlying DETR-based cross-attention architecture at handling temporal grounding tasks.

VI. CONCLUSION

This study’s findings align with observations from the recent DCASE technical reports. The primary objective of this study was not to engineer a novel downstream architecture, but rather to isolate the precise impact of baseline feature extractor selection. As highlighted in retrospective evaluation by the DCASE task organizers, a substantial performance gap remained between top-performing systems and models that focused primarily on feature extraction exploration [14]. This study’s results confirm that selecting a feature extraction model that is compatible with the downstream architecture does influence performance in audio moment retrieval and further fine tuning of the feature representation stage can be beneficial when taken in consideration with what is the optimal feature representation for the task at hand and the decoder framework.

One dataset limitation that should be noted is that due to YouTube videos that were no longer available due to either being taken down or made private, 91 files (approximately 2%) of the CASTELLA dataset was missing for training of M2D-CLAP and OpenFLAM models compared to the original benchmark for MS-CLAP.

Future work should include exploring additional feature extraction strategies, as well as hybrid feature extraction strategies, such as concatenating fine-grained temporal extractors with more global contextual extractors to capture both localized boundary tracking and global acoustic context. In addition, considering the benefits of multi-stage pre-training, incorporating larger and more diverse, temporally focused audio-text datasets for additional pre-training could further enhance task-specific adaptation. To improve retrieval of shorter duration audio moments and tighten boundary resolution, future research should also consider network architectures beyond DETR-based models. As well, based on strengths of DCASE technical reports, future work could include the addition of a frozen LLM final layer for re-ranking to improve recall. Lastly, systematic hyperparameter exploration, such as learning rate drop and IoU loss function weighting, may help to improve fine-resolution localization.

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